

Tide retrospective: the full anharmonic jet (gamma-rung programme, stage 6 of 6)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The gamma-rung programme's finale. Five stages built the analytic ladder — the signed exponential remainder, the second-order rescaled expansions, the moment rates, and the fourth-cumulant limit $t^3\kappa_4 \rightarrow 3\alpha^2/\lambda^5 - \gamma/\lambda^4$. What remained was the recovery reading.

2 The thing formalised

Two declarations in `Laplace/OneD/RecoveryAnharmonicFull.lean`, sorry-free with zero warnings: the observed fourth-cumulant combination and the theorem¹: if the mean susceptibilities, the rescaled covariance susceptibilities, and the fourth-cumulant data of two anharmonic potentials $\frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ eventually agree, then $\lambda_1 = \lambda_2$, $\alpha_1 = \alpha_2$, and $\gamma_1 = \gamma_2$. The germbij note's Theorem 3.1 triangular induction, carried through its third rung: the complete jet of the quartic-truncated potential is determined by observable data, each coefficient read off at its own order.

3 Proof strategy

Composition: the merged two-rung recovery pins (λ, α) ; after substitution, the two fourth-cumulant limits share their $3\alpha^2/\lambda^5$ term, and uniqueness of limits along the eventual equality forces $\gamma_1/\lambda^4 = \gamma_2/\lambda^4$. Fifteen lines of proof for the programme's headline.

4 Roadblocks and resolutions

None: a first-attempt clean build, the programme's cheapest tide. The uniqueness-of-limits recovery pattern, in its fifth use, has zero marginal friction.

5 What was Mathlib, what was new

Mathlib: uniqueness of limits. Seabed: the entire programme. New: the composition — which is the point. A six-stage analytic programme, scoped with a numerically pinned target and a symbolically verified route, ends in a fifteen-line corollary because every hard step was placed where the plan said it would be.

¹`anharmonic_jet_recovery`.

6 Lessons

The gamma rung cost seven tides end to end (stage 2 split at its seam), with the numerical target pinned before stage 1 and never revised. The per-stage pattern — inherit the deliberation, execute against the catalogue, checkpoint WIP commits on the big files — held from the elementary remainder to the finale. Programme-level scoping is the single highest-leverage act: it converted an open-ended “extend Stage 2 to second order” into a sequence of bounded tides.

7 Follow-ups

- Tag the germbij note’s Theorem 3.1 with `anharmonic_jet_recovery` (its jet-recovery instance is now complete for the quartic truncation) and update the documentation surfaces.
- A gamma-programme synthesis retrospective, closing the arc for the mirror’s readers.
- Beyond the programme: the quintic/higher rungs (fifth cumulant onwards) are structurally identical ladders; the multivariate Theorem 3.1 remains a separate commission.